

Internship Evaluation AI- Powered Accounting Chatbot for Ledger Analysis

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📅 Internship Duration : 1st June - 31th July



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📄 Use this agenda to navigate the viva—each section is brief and evidence- focused.

Organization Profile — FinTech Solutions Ltd.

- Founded: 2017 — specializes in accounting automation for SMEs
- Core services: ledger reconciliation, AP/AR automation, analytics
- Clients: 200+ small and medium enterprises across North America
- Role during internship: Data & automation team (product prototyping)

Visual: Company lab environment where prototype testing occurs.





Internship Overview

- Duration: 8 weeks (May–June)
- Mentor: Senior Product Manager — weekly reviews
- Responsibilities: data cleaning, model testing, UI prototyping
- Deliverable: Prototype accounting chatbot for ledger analysis

Focused on practical integration of NLP with accounting workflows.



Project Objectives

- Automate ledger query resolution via conversational interface
- Accelerate ledger analysis and flag anomalies for accountants
- Provide context-aware explanations of transactions and balances
- Integrate with existing ledger CSVs and produce audit- ready logs

- Objectives aligned to reduce manual review time and improve accuracy.

Scope

- Supported inputs: ledger CSVs, journal entries, trial balance
- Primary tasks: query answering, anomaly detection, summary reports
- Limitations: not a substitute for final audit — supports analyst review

Technologies



Python

Data processing and orchestration



NLP Model

Transformer- based intent and entity recognition



SQLite / CSV

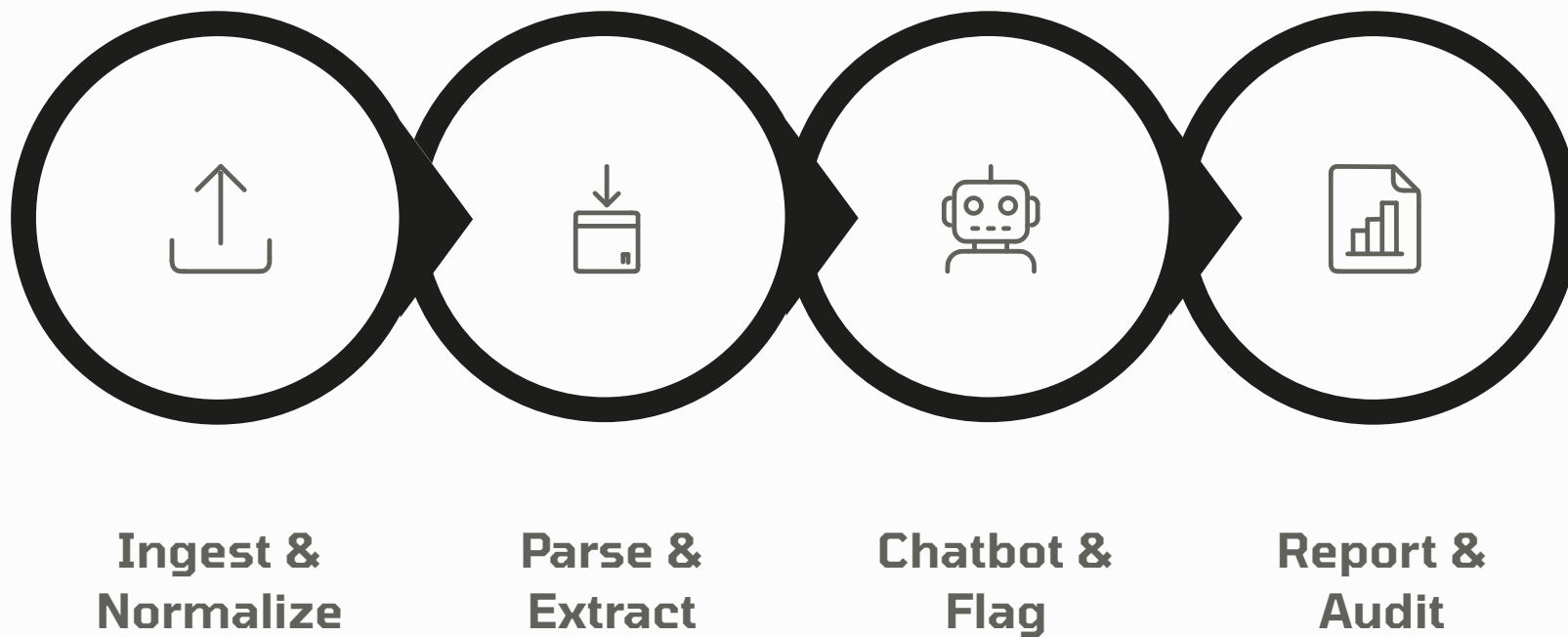
Lightweight ledger storage for prototypes



Web UI

Simple chat interface for demonstrations

System Architecture & Workflow



Pipeline supports repeatable ingestion, conversational QA, anomaly scoring, and export.

Week- Wise Learning Journey – Weeks 1 to 4

Week 1 – Onboarding & Data

Company orientation;
collected sample
ledgers; initial data
cleaning scripts.

Week 2 – Modeling Basics

Explored lightweight NLP
models; built
intent/entity prototypes
on sample entries.

Week 3 – Integration

Linked parser to mock
chat UI; implemented
basic query/response
flows.

Week 4 – Anomaly Detection

Developed rule- based
and statistical checks for
outliers and inconsistent
postings.

Week- Wise Learning Journey — Weeks 5 to 8



Week 5 — Evaluation

Benchmarked responses; measured precision for entity extraction and intent detection.



Week 6 — Refinement

Improved prompt templates; added ledger context windowing and caching.



Week 7 — Compliance & Logging

Implemented audit logs, role checks, and sanitized outputs for PII's.



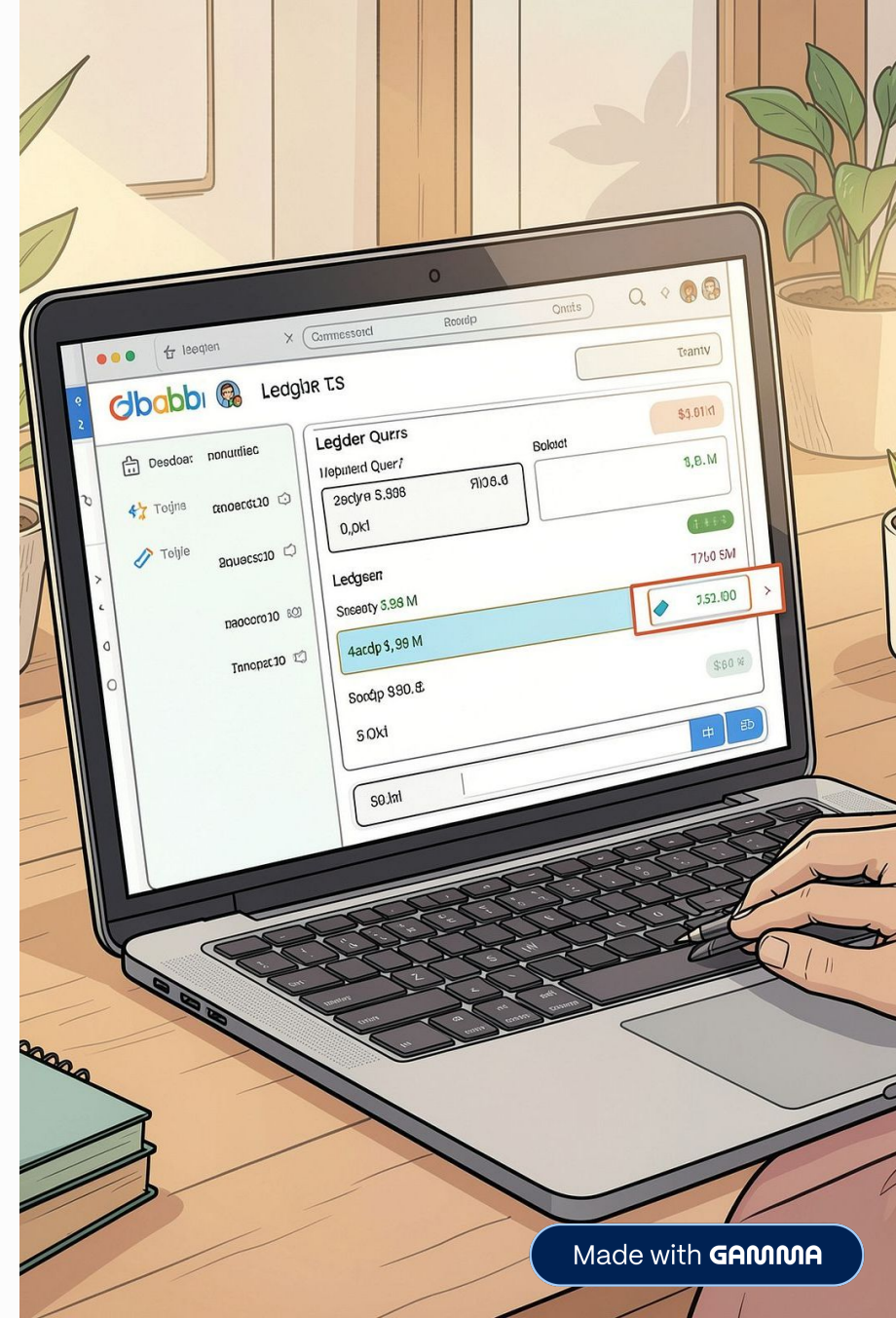
Week 8 — Demo & Documentation

Prepared prototype demo, user guide, and handover notes for the product team.

AI- Powered Accounting Chatbot — Core Features

- Natural language ledger queries (balance, transaction search, trends)
- Anomaly detection: flags duplicates, skewed balances, unexpected postings
- Contextual explanations: plain- language summaries and source links
- Exportable, audit- ready reports and action suggestions for accountants

Designed to assist — not replace — the professional accountant.



Key Results & Metrics

95%

Accuracy

Chatbot answer accuracy on tested ledger queries

92%

Precision

Precision for anomaly detection and flagged transactions

60%

Faster Processing

Reduction in time needed to review common accounting requests

8 hrs

Time Saved

Estimated hours saved per accountant each week

247

Anomalies Detected

Unusual entries, duplicates, and balance issues identified

4.7/5

User Satisfaction

Average satisfaction score from pilot users

Additional outcomes included faster ledger lookup, clearer audit-ready explanations, and a smoother workflow for accountants handling repetitive queries.



Challenges Faced & Key Learnings

Data Inconsistency

Encountered significant challenges with messy and varied ledger data formats, emphasizing the need for robust preprocessing.

NLP Limitations

Fine-tuning general NLP models for niche accounting terminology required iterative refinement and specialized dataset curation.

Integration Complexities

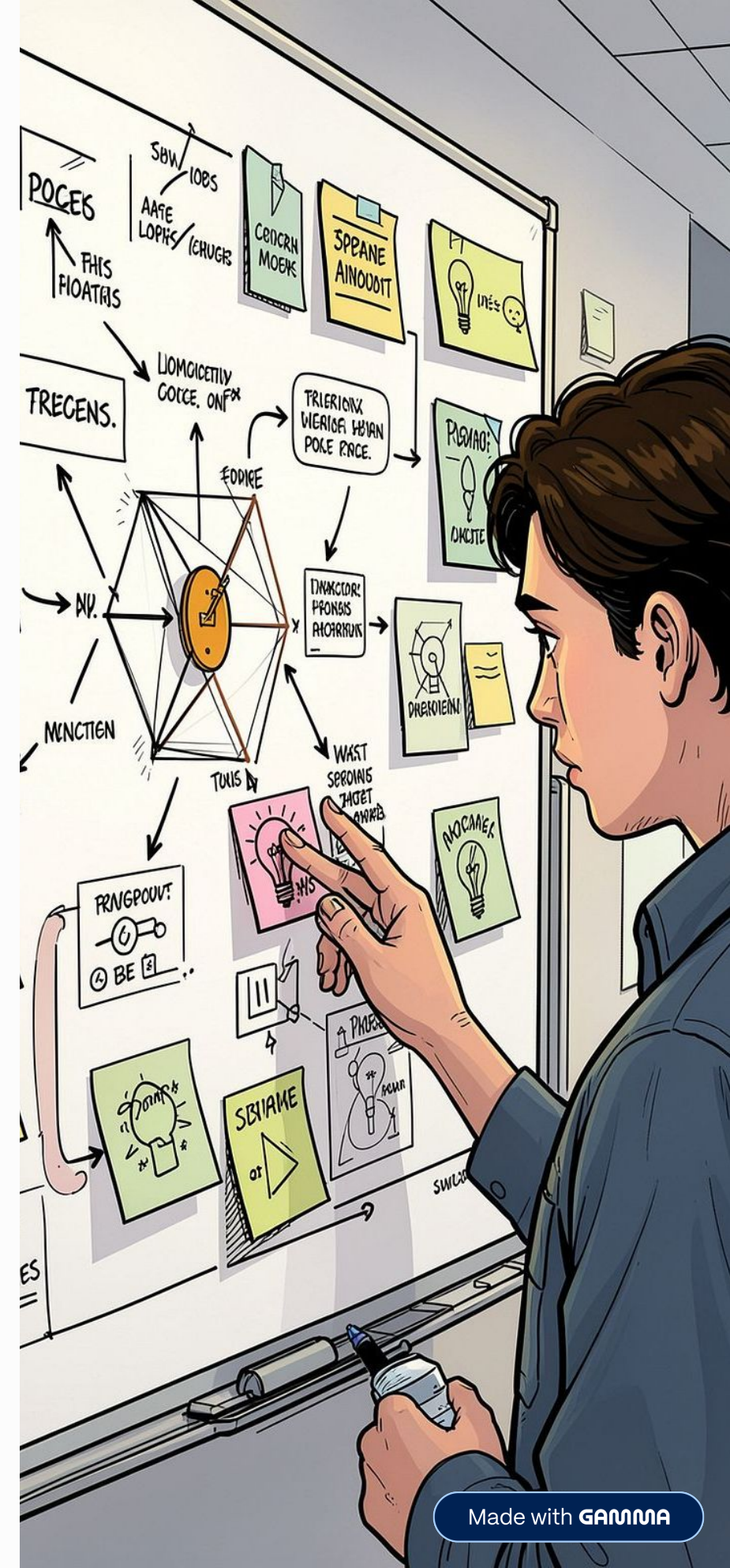
Simulating seamless integration with diverse internal accounting systems highlighted the importance of flexible APIs.

Domain Expertise

Learned the critical value of collaborating with accounting professionals to understand workflow nuances and validation logic.

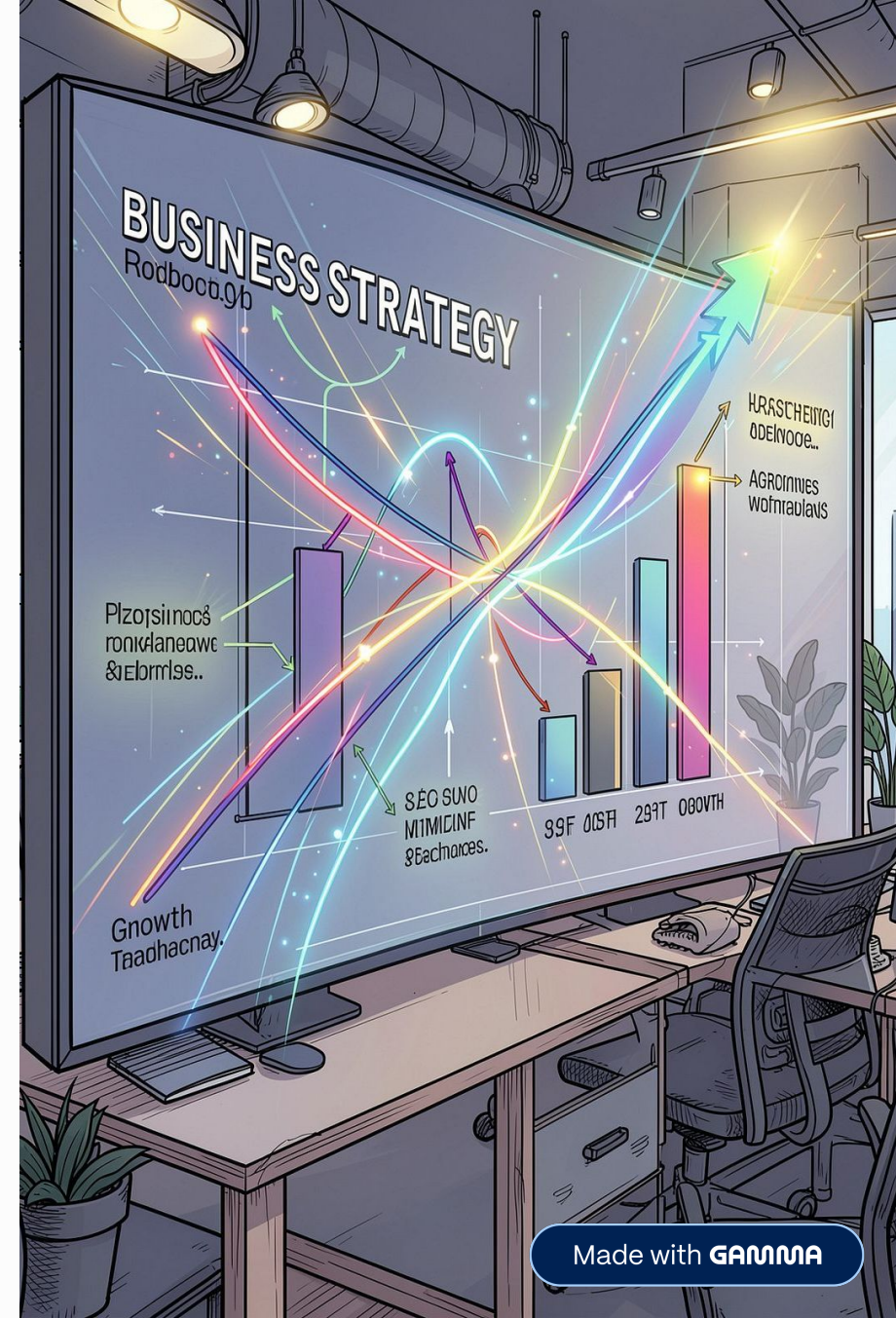
Iterative Development

Adopted an agile approach, realizing that continuous testing and feedback cycles are essential for effective prototyping.



Business Recommendations

- Prioritize **API integrations** with major accounting platforms (e.g., QuickBooks, Xero) for broader market reach.
- Develop **predictive analytics features** to offer proactive financial insights, not just reactive queries.
- Position as a **premium efficiency tool** for mid-market enterprises, highlighting anomaly detection and audit readiness.
- Integrate the chatbot's query engine into existing **FinTech Solutions' product suite** for a unified user experience.



Outcomes, Certification & Conclusion

- **Deliverables:** Successfully prototyped an AI-powered accounting chatbot for ledger analysis.
- **Internship Status:** Completed the 8-week program with a successful project demonstration.
- **Recognition:** Received "Outstanding Intern Project" commendation from FinTech Solutions Ltd.
- **Career Impact:** Gained hands-on experience in AI/NLP, data analysis, and product development within finance.
- **Skills Acquired:** Advanced Python, NLP model refinement, UI/UX prototyping, financial data interpretation.

This project underscored the transformative potential of AI in accounting, offering invaluable experience and direction for future career pursuits in FinTech.

